

Project 67: Voice Emotion Classifier

Goal: Build a system that:

1. Accepts audio input (WAV file or mic)
 2. Extracts features like MFCCs or spectral data
 3. Classifies the speaker's **emotion**
 4. Returns labels like: happy , angry , neutral , sad , etc.
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Popular Datasets

- **RAVDESS:** [Ryerson Audio-Visual Dataset of Emotional Speech](#)
 - **TESS:** Toronto Emotional Speech Set
 - **CREMA-D:** Crowd-sourced Emotional Multimodal Actors Dataset
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Step-by-Step Implementation (Using RAVDESS)

Step 1: Install Required Libraries

```
pip install librosa scikit-learn numpy torch matplotlib
```

Step 2: Feature Extraction – MFCCs

```
import librosa
import numpy as np

def extract_features(file_path):
    y, sr = librosa.load(file_path, duration=3, offset=0.5)
    mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=40)
    mfcc_scaled = np.mean(mfcc.T, axis=0)
    return mfcc_scaled
```

Step 3: Load Data and Labels (RAVDESS Format)

```
import os

emotion_map = {
    '01': 'neutral',
    '02': 'calm',
    '03': 'happy',
    '04': 'sad',
    '05': 'angry',
    '06': 'fearful',
    '07': 'disgust',
    '08': 'surprised'
}

X, y = [], []
for file in os.listdir("ravdess/"):
    if file.endswith(".wav"):
        emotion = emotion_map[file.split("-")[2]]
        feature = extract_features(os.path.join("ravdess", file))
        X.append(feature)
        y.append(emotion)
```

Step 4: Train Classifier (e.g., Random Forest or Simple NN)

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

clf = RandomForestClassifier()
clf.fit(X_train, y_train)

y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred))
```

Step 5: Predict Emotion for New Audio

```
def predict_emotion(file_path):
    feature = extract_features(file_path).reshape(1, -1)
    return clf.predict(feature)[0]

# Try it on a new WAV file
print("Predicted emotion:", predict_emotion("test_speech.wav"))
```

Key Concepts

- **MFCCs**: Capture the timbre and tone from audio—key to detecting emotions
 - **RandomForest** or CNN can be used as classifiers
 - Dataset filenames often encode emotion info (e.g., 03 = happy in RAVDESS)
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Project Summary

- **Input:** Voice file (WAV)
- **Output:** Emotion label (happy , angry , neutral , etc.)
- **Model:** RandomForest trained on MFCCs (or use CNN/LSTM later)
- **Use Case:** Sentiment AI, wellness monitoring, affective agents